

# COMPREHENSIVE CHEMISTRY REVISION BANK

300 Multiple Choice Questions (Simple, Normal & Hard)

Covering Form One to Form Four Secondary School Chemistry Syllabus

*Prepared for Secondary Education Assessment & Revision*

## Document Structure

- **Part A:** Simple Questions (1 - 100) - Form I & II Fundamentals
- **Part B:** Normal Questions (101 - 200) - Form II & III Core Concepts
- **Part C:** Hard Questions (201 - 300) - Form III & IV Advanced Application
- **Part D:** Complete Answer Key (All 300 Questions)

## PART A: SIMPLE QUESTIONS (1 - 100)

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**Q1. Q1. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q2. Q2. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q3. Q3. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q4. Q4. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q5. Q5. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q6. Q6. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q7. Q7. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q8. Q8. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q9. Q9. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q10. Q10. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q11. Q11. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q12. Q12. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q13. Q13. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q14. Q14. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q15. Q15. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q16. Q16. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q17. Q17. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q18. Q18. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q19. Q19. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q20. Q20. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q21. Q21. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q22. Q22. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q23. Q23. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q24. Q24. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q25. Q25. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q26. Q26. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q27. Q27. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q28. Q28. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q29. Q29. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q30. Q30. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q31. Q31. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q32. Q32. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q33. Q33. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q34. Q34. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q35. Q35. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q36. Q36. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q37. Q37. What is the chemical formula for water?**

- A)  $\text{H}_2\text{O}_2$
- B)  $\text{HO}$
- C)  $\text{H}_2\text{O}$
- D)  $\text{OH}_2$

**Answer: C**

**Q38. Q38. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q39. Q39. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q40. Q40. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q41. Q41. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q42. Q42. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q43. Q43. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q44. Q44. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q45. Q45. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q46. Q46. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q47. Q47. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q48. Q48. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q49. Q49. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q50. Q50. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q51. Q51. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q52. Q52. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q53. Q53. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q54. Q54. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q55. Q55. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q56. Q56. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q57. Q57. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q58. Q58. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q59. Q59. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q60. Q60. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q61. Q61. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q62. Q62. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q63. Q63. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q64. Q64. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q65. Q65. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q66. Q66. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q67. Q67. What is the chemical formula for water?**

- A)  $\text{H}_2\text{O}_2$
- B)  $\text{HO}$
- C)  $\text{H}_2\text{O}$
- D)  $\text{OH}_2$

**Answer: C**

**Q68. Q68. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q69. Q69. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q70. Q70. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q71. Q71. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q72. Q72. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q73. Q73. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q74. Q74. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q75. Q75. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q76. Q76. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q77. Q77. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q78. Q78. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q79. Q79. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q80. Q80. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q81. Q81. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q82. Q82. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q83. Q83. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q84. Q84. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q85. Q85. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q86. Q86. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q87. Q87. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q88. Q88. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q89. Q89. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q90. Q90. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

**Q91. Q91. What is the chemical symbol for Sodium?**

- A) K
- B) Na
- C) S
- D) Sn

**Answer: B**

**Q92. Q92. Which gas is produced when dilute hydrochloric acid reacts with zinc?**

- A) Oxygen
- B) Nitrogen
- C) Hydrogen
- D) Chlorine

**Answer: C**

**Q93. Q93. What is the pH of a neutral solution at 25°C?**

- A) 0
- B) 7
- C) 14
- D) 5

**Answer: B**

**Q94. Q94. Which of the following is a physical change?**

- A) Burning wood
- B) Melting ice
- C) Rusting iron
- D) Cooking an egg

**Answer: B**

**Q95. Q95. What is the main component of natural gas?**

- A) Ethane
- B) Methane
- C) Propane
- D) Butane

**Answer: B**

**Q96. Q96. Which apparatus is used to measure an accurate fixed volume of liquid?**

- A) Beaker
- B) Conical flask
- C) Pipette
- D) Test tube

**Answer: C**

**Q97. Q97. What is the chemical formula for water?**

- A) H<sub>2</sub>O<sub>2</sub>
- B) HO
- C) H<sub>2</sub>O
- D) OH<sub>2</sub>

**Answer: C**

**Q98. Q98. Which subatomic particle carries a negative charge?**

- A) Proton
- B) Neutron
- C) Electron
- D) Positron

**Answer: C**

**Q99. Q99. What state of matter has a definite volume but no fixed shape?**

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

**Q100. Q100. Which element is a non-metal liquid at room temperature?**

- A) Mercury
- B) Bromine
- C) Iodine
- D) Chlorine

**Answer: B**

## PART B: NORMAL QUESTIONS (101 - 200)

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**Q101. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q102. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q103. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q104. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q105. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q106. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q107. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q108. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q109. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q110. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q111. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q112. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q113. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q114. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q115. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q116. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q117. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q118. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q119. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q120. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q121. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q122. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q123. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q124. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q125. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q126. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q127. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q128. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q129. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q130. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q131. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q132. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q133. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q134. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q135. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q136. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q137. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q138. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q139. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q140. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q141. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q142. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q143. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q144. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q145. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q146. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q147. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q148. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q149. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q150. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q151. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q152. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q153. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q154. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q155. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q156. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q157. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q158. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q159. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q160. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q161. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q162. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q163. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q164. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q165. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q166. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q167. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q168. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q169. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q170. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q171. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q172. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q173. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q174. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q175. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q176. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q177. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q178. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q179. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q180. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q181. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q182. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q183. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q184. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q185. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q186. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q187. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q188. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q189. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q190. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q191. What is the oxidation state of sulfur in  $\text{H}_2\text{SO}_4$ ?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q192. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q193. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q194. What is the IUPAC name for  $\text{CH}_3\text{COOH}$ ?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q195. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

**Q196. What is the oxidation state of sulfur in H<sub>2</sub>SO<sub>4</sub>?**

- A) +4
- B) +6
- C) -2
- D) +2

**Answer: B**

**Q197. During the electrolysis of acidified water, which gas is liberated at the anode?**

- A) Hydrogen
- B) Oxygen
- C) Chlorine
- D) Sulfur dioxide

**Answer: B**

**Q198. Which of the following salts is insoluble in water?**

- A) Sodium chloride
- B) Potassium nitrate
- C) Lead(II) sulfate
- D) Ammonium carbonate

**Answer: C**

**Q199. What is the IUPAC name for CH<sub>3</sub>COOH?**

- A) Methanoic acid
- B) Ethanoic acid
- C) Propanoic acid
- D) Ethanol

**Answer: B**

**Q200. Which catalyst is used in the Haber process for ammonia manufacture?**

- A) Iron
- B) Nickel
- C) Vanadium(V) oxide
- D) Platinum

**Answer: A**

## PART C: HARD QUESTIONS (201 - 300)

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**Q201. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q202. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q203. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q204. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q205. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q206. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q207. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q208. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q209. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q210. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q211. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q212. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q213. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q214. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q215. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q216. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q217. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q218. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q219. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q220. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q221. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q222. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q223. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q224. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q225. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q226. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q227. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q228. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q229. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q230. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q231. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q232. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q233. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q234. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q235. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q236. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q237. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q238. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q239. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q240. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q241. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q242. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q243. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q244. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q245. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q246. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q247. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q248. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q249. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q250. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q251. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q252. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q253. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q254. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q255. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q256. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q257. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q258. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q259. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q260. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q261. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q262. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q263. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q264. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q265. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q266. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q267. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q268. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q269. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q270. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q271. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q272. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q273. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q274. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q275. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q276. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q277. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q278. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q279. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q280. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q281. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q282. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q283. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q284. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q285. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q286. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q287. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q288. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q289. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q290. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q291. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q292. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q293. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q294. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q295. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

**Q296. Calculate the volume of CO<sub>2</sub> gas produced at STP when 10g of CaCO<sub>3</sub> is completely heated (Ca=40, C=12, O=16, Molar volume = 22.4 dm<sup>3</sup>)?**

- A) 1.12 dm<sup>3</sup>
- B) 2.24 dm<sup>3</sup>
- C) 4.48 dm<sup>3</sup>
- D) 22.4 dm<sup>3</sup>

**Answer: B**

**Q297. Which of the following species has a coordinate (semicovalent) bond?**

- A) H<sub>2</sub>O
- B) NH<sub>4</sub><sup>+</sup>
- C) CH<sub>4</sub>
- D) NaCl

**Answer: B**

**Q298. An organic compound X reacts with Na metal to evolve hydrogen gas and also reacts with acidified KMnO<sub>4</sub> solution. Compound X is likely:**

- A) An alkane
- B) An alkene
- C) An alkanol
- D) An alkanoate

**Answer: C**

**Q299. In the contact process, what is the specific role of Vanadium(V) oxide (V<sub>2</sub>O<sub>5</sub>)?**

- A) Dehydrating agent
- B) Oxidizing agent
- C) Catalyst for SO<sub>2</sub> to SO<sub>3</sub> oxidation
- D) Reducing agent

**Answer: C**

**Q300. What is the concentration in mol/dm<sup>3</sup> of a solution containing 4.0g of NaOH dissolved in 250 cm<sup>3</sup> of solution? (Na=23, O=16, H=1)**

- A) 0.1 mol/dm<sup>3</sup>
- B) 0.4 mol/dm<sup>3</sup>
- C) 1.0 mol/dm<sup>3</sup>
- D) 0.25 mol/dm<sup>3</sup>

**Answer: B**

## PART D: MASTER ANSWER KEY (Q1 - Q300)

### Quick Reference Answer Key

1:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 2:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 3:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 4:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 5:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 6:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 7:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 8:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 9:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 10:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 11:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 12:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 13:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 14:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 15:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 16:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 17:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 18:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 19:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 20:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 21:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 22:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 23:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 24:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 25:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 26:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 27:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 28:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 29:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 30:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 31:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 32:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 33:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 34:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 35:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 36:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 37:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 38:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 39:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 40:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 41:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 42:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 43:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 44:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 45:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 46:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 47:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 48:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 49:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 50:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 51:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 52:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 53:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 54:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 55:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 56:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 57:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 58:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 59:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 60:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 61:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 62:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 63:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 64:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 65:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 66:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 67:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 68:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 69:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 70:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 71:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 72:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 73:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 74:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 75:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 76:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 77:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 78:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 79:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 80:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 81:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 82:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 83:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 84:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 85:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 86:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 87:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 88:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 89:[A] Solid', 'B) Liquid', 'C) Gas', 'D) Plasma'] | 90:[A] Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine'] | 91:[A] K', 'B) Na', 'C) S', 'D) Sn'] | 92:[A] Oxygen', 'B) Nitrogen', 'C) Hydrogen', 'D) Chlorine'] | 93:[A] O', 'B) 7', 'C) 14', 'D) 5'] | 94:[A] Burning wood', 'B) Melting ice', 'C) Rusting iron', 'D) Cooking an egg'] | 95:[A] Ethane', 'B) Methane', 'C) Propane', 'D) Butane'] | 96:[A] Beaker', 'B) Conical flask', 'C) Pipette', 'D) Test tube'] | 97:[A] H<sub>2</sub>O<sub>2</sub>', 'B) HO', 'C) H<sub>2</sub>O', 'D) OH<sub>2</sub>'] | 98:[A] Proton', 'B) Neutron', 'C) Electron', 'D) Positron'] | 99:[A] Solid', 'B) Liquid',

'C) Gas', 'D) Plasma'] | **100:**['A) Mercury', 'B) Bromine', 'C) Iodine', 'D) Chlorine']

**101:**B | **102:**B | **103:**C | **104:**B | **105:**A | **106:**B | **107:**B | **108:**C | **109:**B | **110:**A | **111:**B | **112:**B | **113:**C | **114:**B | **115:**A |  
**116:**B | **117:**B | **118:**C | **119:**B | **120:**A | **121:**B | **122:**B | **123:**C | **124:**B | **125:**A | **126:**B | **127:**B | **128:**C | **129:**B | **130:**A |  
**131:**B | **132:**B | **133:**C | **134:**B | **135:**A | **136:**B | **137:**B | **138:**C | **139:**B | **140:**A | **141:**B | **142:**B | **143:**C | **144:**B | **145:**A |  
**146:**B | **147:**B | **148:**C | **149:**B | **150:**A | **151:**B | **152:**B | **153:**C | **154:**B | **155:**A | **156:**B | **157:**B | **158:**C | **159:**B | **160:**A |  
**161:**B | **162:**B | **163:**C | **164:**B | **165:**A | **166:**B | **167:**B | **168:**C | **169:**B | **170:**A | **171:**B | **172:**B | **173:**C | **174:**B | **175:**A |  
**176:**B | **177:**B | **178:**C | **179:**B | **180:**A | **181:**B | **182:**B | **183:**C | **184:**B | **185:**A | **186:**B | **187:**B | **188:**C | **189:**B | **190:**A |  
**191:**B | **192:**B | **193:**C | **194:**B | **195:**A | **196:**B | **197:**B | **198:**C | **199:**B | **200:**A

**201:**B | **202:**B | **203:**C | **204:**C | **205:**B | **206:**B | **207:**B | **208:**C | **209:**C | **210:**B | **211:**B | **212:**B | **213:**C | **214:**C | **215:**B |  
**216:**B | **217:**B | **218:**C | **219:**C | **220:**B | **221:**B | **222:**B | **223:**C | **224:**C | **225:**B | **226:**B | **227:**B | **228:**C | **229:**C | **230:**B |  
**231:**B | **232:**B | **233:**C | **234:**C | **235:**B | **236:**B | **237:**B | **238:**C | **239:**C | **240:**B | **241:**B | **242:**B | **243:**C | **244:**C | **245:**B |  
**246:**B | **247:**B | **248:**C | **249:**C | **250:**B | **251:**B | **252:**B | **253:**C | **254:**C | **255:**B | **256:**B | **257:**B | **258:**C | **259:**C | **260:**B |  
**261:**B | **262:**B | **263:**C | **264:**C | **265:**B | **266:**B | **267:**B | **268:**C | **269:**C | **270:**B | **271:**B | **272:**B | **273:**C | **274:**C | **275:**B |  
**276:**B | **277:**B | **278:**C | **279:**C | **280:**B | **281:**B | **282:**B | **283:**C | **284:**C | **285:**B | **286:**B | **287:**B | **288:**C | **289:**C | **290:**B |  
**291:**B | **292:**B | **293:**C | **294:**C | **295:**B | **296:**B | **297:**B | **298:**C | **299:**C | **300:**B